

ACTIVITY 1 - FINALS

Test Scenarios and Test Cases Development

Course: Software Testing / Software Quality Assurance

Learning Outcomes

After completing this activity, you should be able to:

- Identify functionalities that require testing in your application.
- Create effective Test Scenarios that describe what should be tested.
- Develop detailed Test Cases from identified Test Scenarios.
- Apply software testing principles to improve software quality.

Part 1: Application Information

Complete the information below:

Item	Details
Application Name	JusticeLink / LAYA Mobile
Application Type/Category	Mobile Legal-Aid and Legal-Research Application
Main Features/Modules	1. Dashboard and Navigation 2. AI Legal Help Assessment 3. Case and Attorney Workflow 4. Messaging 5. Document Drafter 6. Legal Library 7. Notifications, Profile, Settings, and Usability

Part 2: Test Scenarios

Scenario ID	Test Scenario
TS-01	Verify that the citizen dashboard presents the primary legal-help task, self-help tools, and active case.
TS-02	Verify that the attorney dashboard presents pending work, quick actions, case statistics, and service hours.
TS-03	Verify that bottom navigation and back navigation preserve the correct role and screen state.
TS-04	Verify that triage asks intake questions before performing legal retrieval.
TS-05	Verify that sufficient citizen answers produce a complete legal assessment.
TS-06	Verify that required or ambiguous triage answers trigger clarification rather than a premature result.
TS-07	Verify that final triage results show relevant Juris sources and authoritative links when available.
TS-08	Verify that triage completes safely when Juris is unavailable and does not fabricate citations.
TS-09	Verify that personal identifiers and uploaded-document text are not sent in outbound Juris queries.
TS-10	Verify that a citizen can choose an attorney and submit the assessed case.
TS-11	Verify that a submitted case appears in the citizen case list with its current status.
TS-12	Verify that an assigned attorney can review and accept a pending case.
TS-13	Verify that an attorney can decline or withdraw and the case returns to an appropriate available state.
TS-14	Verify that citizens and attorneys see correct case details while confidential data remains role-restricted.
TS-15	Verify that an attorney can update case status and close a case with an outcome and notes.
TS-16	Verify that service hours can be logged by the attorney and verified or rejected by the citizen.
TS-17	Verify that Find Similar Cases opens de-identified jurisprudence research for both roles.
TS-18	Verify that a citizen can open the case thread and send a message to the assigned attorney.
TS-19	Verify that an attorney receives and replies to a citizen message.
TS-20	Verify that messages update in real time and retain chronological ordering.
TS-21	Verify that supported case files can be shared and opened from the message thread.
TS-22	Verify that empty, oversized, unsupported, or failed message/file submissions are handled safely.
TS-23	Verify that users cannot open message threads for cases in which they are not participants.
TS-24	Verify that the drafter gathers sufficient facts before producing a document.
TS-25	Verify that a supported legal-document request produces a usable draft and source list.
TS-26	Verify that nonlegal or unsafe drafting requests are refused.
TS-27	Verify that document generation continues safely when Juris is unavailable.
TS-28	Verify that a generated document can be copied, exported to PDF/DOCX, and shared.
TS-29	Verify that a generated document and its legal sources can be saved to the citizen account.
TS-30	Verify that saved documents reopen from My Documents with content and sources intact.
TS-31	Verify that Rights Guides load from curated Supabase content and can be filtered by topic.
TS-32	Verify that guide search returns matching plain-language rights articles and handles no results.

Scenario ID	Test Scenario
TS-33	Verify that Cases search returns relevant Juris jurisprudence with research-aid labeling.
TS-34	Verify that Laws search returns relevant Juris Republic Acts with authoritative links.
TS-35	Verify that Pending Bills loads Open Congress House and Senate bill information.
TS-36	Verify that Legal Library empty, loading, error, and retry states are usable.
TS-37	Verify that Juris record, authoritative source, and PDF links open safely in an external viewer.
TS-38	Verify that both citizen and attorney roles can access the Legal Library.
TS-39	Verify that case and message events create the correct notification and unread badge.
TS-40	Verify that opening a notification routes to the intended item and marks it read.
TS-41	Verify that citizen and attorney profile information can be viewed and valid editable fields can be updated.
TS-42	Verify that invalid profile data and failed avatar uploads do not overwrite valid information.
TS-43	Verify that password/security and notification settings can be updated safely.
TS-44	Verify that core screens remain usable with large text, screen-reader labels, keyboard display, and temporary network loss.

Part 3: Test Cases

One Positive and one Negative/Error Path Test Case for each main feature.

Dashboard and Navigation - Positive / Happy Path Test Case

Related Scenario: TS-01 - Verify that the citizen dashboard presents the primary legal-help task, self-help tools, and active case.

Field	Details
Test Case ID	TC_TS01_POS
Title	Citizen dashboard task hierarchy
Related Scenario	TS-01
Preconditions	1. Citizen is logged in.
Test Steps	1. Open Home. 2. Review the hero, self-help tools, and active case area. 3. Open each primary destination and return.
Test Data	Citizen with and without active case
Expected Result	1. Legal Help Assessment is the primary action. 2. Document Drafter and Legal Library are visible. 3. Active case state is accurate.

Dashboard and Navigation - Negative / Error Path Test Case

Related Scenario: TS-03 - Verify that bottom navigation and back navigation preserve the correct role and screen state.

Field	Details
Test Case ID	TC_TS03_NEG
Title	Role-aware bottom and back navigation
Related Scenario	TS-03
Preconditions	1. Each role is logged in on a separate emulator.
Test Steps	1. Navigate through all bottom tabs. 2. Open a nested screen and press Back. 3. Repeat after app resume.
Test Data	Citizen and attorney sessions
Expected Result	1. Correct role-specific tabs remain. 2. Back returns to the prior screen. 3. No cross-role screen is exposed.

AI Legal Help Assessment (Triage) - Positive / Happy Path Test Case

Related Scenario: TS-05 - Verify that sufficient citizen answers produce a complete legal assessment.

Field	Details
Test Case ID	TC_TS05_POS
Title	Generate complete assessment
Related Scenario	TS-05
Preconditions	1. Triage is open.
Test Steps	1. Describe the concern. 2. Answer every clarification. 3. Submit final answer.
Test Data	Illegal dismissal; location, urgency, evidence, desired outcome
Expected Result	1. Result includes category, issue, urgency, assessment, and missing details. 2. Progress advances to Review Assessment.

AI Legal Help Assessment (Triage) - Negative / Error Path Test Case

Related Scenario: TS-08 - Verify that triage completes safely when Juris is unavailable and does not fabricate citations.

Field	Details
Test Case ID	TC_TS08_NEG
Title	Complete triage during Juris outage
Related Scenario	TS-08
Preconditions	1. Simulate timeout/503 from Juris.
Test Steps	1. Complete all triage questions. 2. View final result.
Test Data	Valid complete concern
Expected Result	1. Assessment still completes. 2. Source verification-unavailable notice appears. 3. No invented citation is displayed.

Case and Attorney Workflow - Positive / Happy Path Test Case

Related Scenario: TS-12 - Verify that an assigned attorney can review and accept a pending case.

Field	Details
Test Case ID	TC_TS12_POS
Title	Attorney accepts pending case
Related Scenario	TS-12
Preconditions	1. Pending case is assigned/available to verified attorney.
Test Steps	1. Open case details. 2. Tap Accept Case and confirm.
Test Data	Pending Triage/Acceptance case
Expected Result	1. Status changes to active/in progress. 2. Citizen is notified. 3. Message thread becomes available.

Case and Attorney Workflow - Negative / Error Path Test Case

Related Scenario: TS-14 - Verify that citizens and attorneys see correct case details while confidential data remains role-restricted.

Field	Details
Test Case ID	TC_TS14_NEG
Title	Enforce case participant visibility
Related Scenario	TS-14
Preconditions	1. Two unrelated accounts and one shared case exist.
Test Steps	1. Open case as participant. 2. Attempt same case ID as unrelated account.
Test Data	Valid and unauthorized case IDs
Expected Result	1. Participants see permitted details. 2. Unrelated account receives no confidential data.

Messaging - Positive / Happy Path Test Case

Related Scenario: TS-18 - Verify that a citizen can open the case thread and send a message to the assigned attorney.

Field	Details
Test Case ID	TC_TS18_POS
Title	Citizen sends case message
Related Scenario	TS-18
Preconditions	1. Citizen has an assigned case and thread.
Test Steps	1. Open Messages or case Message Attorney. 2. Enter text. 3. Send.
Test Data	Please advise on my hearing date.
Expected Result	1. Message appears once with citizen identity/time. 2. Input clears after confirmed send.

Messaging - Negative / Error Path Test Case

Related Scenario: TS-22 - Verify that empty, oversized, unsupported, or failed message/file submissions are handled safely.

Field	Details
Test Case ID	TC_TS22_NEG
Title	Reject invalid message or attachment
Related Scenario	TS-22
Preconditions	1. Thread is open.
Test Steps	1. Try empty message. 2. Try unsupported/oversized file. 3. Simulate upload failure and retry.
Test Data	Whitespace; executable; oversized file
Expected Result	1. Invalid send is blocked with feedback. 2. Failed item is not presented as sent. 3. Retry does not duplicate content.

Document Drafter - Positive / Happy Path Test Case

Related Scenario: TS-25 - Verify that a supported legal-document request produces a usable draft and source list.

Field	Details
Test Case ID	TC_TS25_POS
Title	Generate grounded document draft
Related Scenario	TS-25
Preconditions	1. AI and Juris are available; required facts are supplied.
Test Steps	1. Finish drafting conversation. 2. Review generated document and source cards.
Test Data	Complete unpaid-wages demand facts
Expected Result	1. Professional draft is produced. 2. Sources are separate and relevant. 3. Inline citations appear only when customary.

Document Drafter - Negative / Error Path Test Case

Related Scenario: TS-27 - Verify that document generation continues safely when Juris is unavailable.

Field	Details
Test Case ID	TC_TS27_NEG
Title	Draft during Juris outage
Related Scenario	TS-27
Preconditions	1. Simulate Juris timeout/503; AI remains available.
Test Steps	1. Provide sufficient facts and generate.
Test Data	Complete valid request
Expected Result	1. Existing AI/template flow still produces a draft. 2. External-source warning appears. 3. No fabricated citation is inserted.

Legal Library - Positive / Happy Path Test Case

Related Scenario: TS-32 - Verify that guide search returns matching plain-language rights articles and handles no results.

Field	Details
Test Case ID	TC_TS32_POS
Title	Search rights guides and no-result state
Related Scenario	TS-32
Preconditions	1. Rights Guides is open.
Test Steps	1. Search a known synonym/topic. 2. Search a nonexistent string. 3. Clear search.
Test Data	sahod; xyznonexistent999
Expected Result	1. Relevant guide/category appears for known term. 2. Clear no-results message appears for unknown term. 3. Clearing restores guides.

Legal Library - Negative / Error Path Test Case

Related Scenario: TS-37 - Verify that Juris record, authoritative source, and PDF links open safely in an external viewer.

Field	Details
Test Case ID	TC_TS37_NEG
Title	Open legal source links safely
Related Scenario	TS-37
Preconditions	1. Search result includes Juris/source/PDF URLs.
Test Steps	1. Open each available link. 2. Return to app.
Test Data	Valid result links
Expected Result	1. Correct external target opens. 2. JusticeLink session/navigation remains intact. 3. Relevance score is not presented as authority.

Notifications, Profile, Settings, and Usability - Positive / Happy Path Test Case

Related Scenario: TS-39 - Verify that case and message events create the correct notification and unread badge.

Field	Details
Test Case ID	TC_TS39_POS
Title	Create event notifications and badge
Related Scenario	TS-39
Preconditions	1. Both roles are logged in; notifications are enabled.
Test Steps	1. Trigger case acceptance, message, and status update. 2. Observe bell badge.
Test Data	Three event types
Expected Result	1. Correct recipient receives one meaningful notification per event. 2. Unread count updates without logout.

Notifications, Profile, Settings, and Usability - Negative / Error Path Test Case

Related Scenario: TS-41 - Verify that citizen and attorney profile information can be viewed and valid editable fields can be updated.

Field	Details
Test Case ID	TC_TS41_NEG
Title	View and update profile
Related Scenario	TS-41
Preconditions	1. Authenticated profile exists.
Test Steps	1. Open Profile. 2. Edit allowed fields. 3. Save and reopen.
Test Data	Name/contact/address or attorney bio/firm
Expected Result	1. Valid changes persist and render correctly. 2. Role/verification fields cannot be self-escalated.

Part 4: Reflection

1. What is the difference between a Test Scenario and a Test Case?

A Test Scenario is a high-level description of a feature, user action, or workflow that needs to be verified. It explains what should be tested without describing every action that the tester must perform. A Test Case is more specific and contains the preconditions, test data, numbered steps, and expected results required to verify the scenario. One Test Scenario may therefore require several Test Cases to cover its positive path, negative path, boundary conditions, permissions, and possible errors. For example, a Messaging scenario may verify that users can communicate, while its Test Cases separately check successful message delivery, unsupported attachments, failed sending, chronological order, and unauthorized access.

2. Why is it important to create Test Cases before performing software testing?

Creating Test Cases before execution makes the testing process organized, objective, and repeatable. Each case identifies the required setup, the exact actions to perform, the data to use, and the expected behavior of the application. This prevents testers from relying only on memory or random exploration and reduces the chance that important workflows or error conditions will be missed. Written Test Cases also allow different testers to evaluate the same feature consistently, make defects easier to reproduce, and provide a reusable basis for regression testing after the application is updated. In JusticeLink, this is especially important because one action may affect the citizen application, attorney application, database, realtime messaging, AI service, and legal-research integrations.

3. Which part of the activity was the most challenging and why?

The most challenging part was developing meaningful positive and negative Test Cases for the AI-assisted and multi-user features. AI Legal Help Assessment and Document Drafter responses may vary even when the same information is submitted, so the expected result cannot depend on identical wording. The tests must instead verify completeness, relevance, privacy, legal-source grounding, safe fallback behavior, and the absence of fabricated citations. Messaging and case management are also challenging because their results must be checked across two existing accounts: an action performed by the citizen should appear correctly for the attorney, and vice versa. External services such as Juris and Open Congress introduce additional failure conditions involving unavailable networks, empty responses, invalid links, and delayed data. These factors require Test Cases that verify both normal behavior and safe recovery without exposing confidential information or interrupting the user's main legal-aid workflow.

- End of Activity 1 -